MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 304TL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Mounting position: Vertical, base up or down

Weight: 9-ounces (255.15 grams) nominal

ABSOLUTE RATINGS: C Telegraphy

Parameter:	Fl	Ef	Eb	Ec	Ib	Pg	Pp
Unit:	MHz	V	Vdc	Vdc	mAdc	W	W
Maximum:	8	5.25 or 10.5	3,000	0	900	50	300
Minimum:	---	4.75 or 9.5	---	---	---	---	---
TEST CONDITIONS:	---	5.0 Vac	1,000	Adjust	300	---	---

GENERAL:

Qualification - Required

Ⓕ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Peak emission	-	eb = ec = 2,500 v	0.65	II	is	10.0	---	a
1261	Electrode voltage (1) (grid)	-		0.65	II	Ec(1)	-38.0	-50.0	Vdc
1261	Electrode voltage (2) (grid)	-	Eb = 14 kVdc; Ec/Ib = 1.0 mAdc	0.65	II	Ec(2)	---	-1,750	Vdc
1266	Total grid current	1		0.65	II	Ic	---	-25	µAdc
1266	Primary grid emission	- -	Ic = 265 mAdc; t = 15 sec; Ef = 5.5 Vac; anode floating	0.65	II	Isg	---	-500	µAdc
1301	Filament current	-	Connect in parallel	0.65	II	If	24.0	28.0	Aac
	Quality conformance inspection, part 2								
1236	Power oscillation	-	Eb = 3,000 Vdc; Ec/Ib = 250 mAdc; F = 8 MHz; Ic = 45 mAdc; Rg = 10,000 ohms	---	---	Po	450	---	W (useful)
1316	Amplification factor	-		---	---	Mu	10.5	12.5	---
1331	Direct-interelectrode capacitance	-		---	---	Cgp	7.1	10.2	pF
						Cin	10.0	14.3	pF
						Cout	---	0.9	pF
	Quality conformance inspection, part 3								
(F) 1031	Low-frequency vibration	2	No voltages applied	---	---	---	---	---	---
(F) 1036	Bump	2	Angle = 10°	---	---	---	---	---	---
---	Life test	-	Group C; power osciallation t = 500 hours	---	--	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	8.0	---	a
1266	Primary grid emission	-		---	---	Isg	---	-1.0	mAdc

NOTES:

1. This test shall be the first test perfromed at the conclusion of the holding period.
2. (F) This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is proudction. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

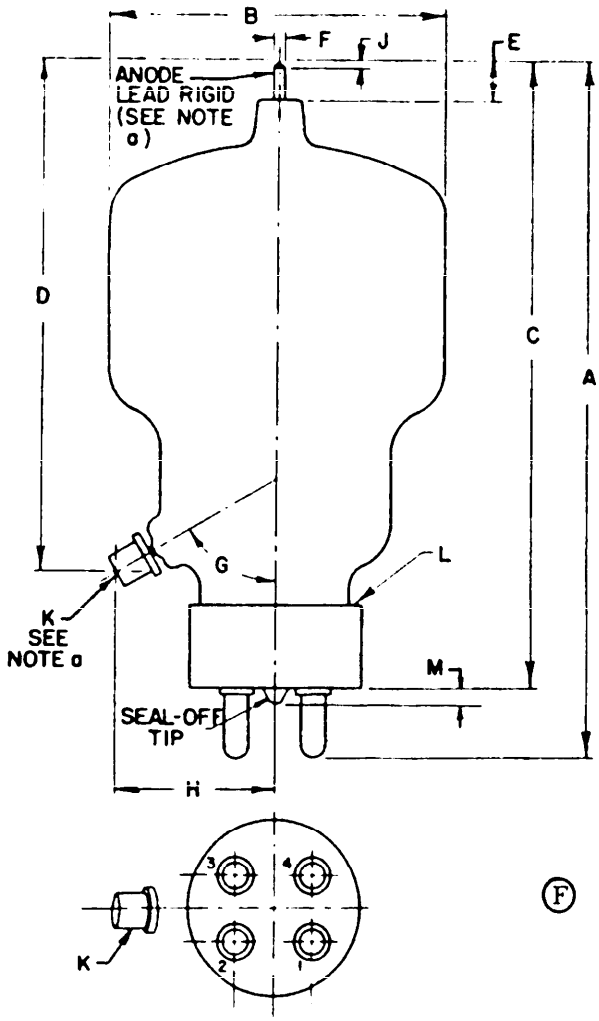
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2940)

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PIN CONNECTIONS	
Pin No.	Element
1	f1
2	f1
3	f2
4	f2
top lead	a
cap	g

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	7.250 (184.15)	7.625 (193.68)
B	---	3.563 (90.50)
C	6.500 (165.10)	6.875 (174.63)
D	5.313 (134.95)	5.688 (144.48)
E	.438 (11.13)	---
H	1.688 (42.88)	1.813 (46.05)
Quality conformance inspection, part 3 (see note c)		
F	.120 (3.05)	.126 (3.20)
K	Cap: C1-23 (EIA)	
L	Base: A4-62 (EIA)	
M	---	.187 (4.75)
Reference dimensions		
G	60°	
J	.031 X 45° or .005R (.79 X 45° or .13R)	

NOTES:

a. Heat radiating connectors HR-6 (grid) and HR-7 (anode) or equivalent, are recommended for all applications.

b. Dimension A includes .060 (1.52 mm) solder tolerance on length of pins.

c. These dimensions shall be checked annually with the following sampling plan:

n1 = 4 c1 = 0 where c2 represents the total allowable failures
n2 = 4 c2 = 1; for the first and second samples combined.

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 basis.

FIGURE 1. Outline drawing of electron tube type 304TL.

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WASHINGTON, D. C. 20360

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